The Quantum Optimization Challenge

How can quantum computing be used to address a meaningful real-world optimization problem with competing objectives or limited resources?

Challenge brief

Real-world decisions involve choosing the best option while working with limited time, resources, or capacity. Teams must select ONE of the six application domains below, identify their own meaningful optimization problem, and develop a quantum or hybrid quantum-classical approach using IBM Qiskit. There is no pre-assigned problem: each team defines its own use case within the chosen domain.

What each team must do

01 Identify — a concrete, relevant optimization problem and its intended users or context.

02 Formulate — the decision variables, objective function, constraints, and a small test instance.

03 Justify — the selected quantum algorithm and show how the problem is represented in it.

04 Implement — and run a meaningful quantum component in Qiskit can be a tool or a subpart.

05 Evaluate — the result using a suitable classical, exact, or analytically verifiable reference.

06 Reflect — on correctness, feasibility, practical limitations, and any trade-offs observed.

Choose one of six application domains

Select one domain and define your own optimization problem. The examples below are only to help you get started; they are not fixed problem statements. Use a small, simulator-friendly instance. Physical hardware and demonstrated quantum speedup are not required.

ELIGIBLE DOMAIN	ILLUSTRATIVE OPTIMIZATION TASKS
1. UAVs & Robotics	Mission task allocation, inspection scheduling, or robot charging schedules
2. Disaster Response	Allocation of rescue teams, emergency supplies, or shelter resources
3. Healthcare	Patient appointment scheduling or allocation of limited medical equipment
4. Agriculture	Irrigation scheduling or assignment of equipment across fields
5. Logistics & Transportation	Delivery assignments, vehicle scheduling, or distribution planning
6. Energy & Smart Infrastructure	Power-use scheduling or allocation of charging slots

Deliverables

- A concise problem statement with objective, constraints, and motivation.
- Working Qiskit code and a reproducible demonstration on at least one small instance provide Github Repositories link.
- A brief report containing Architecture diagram ,results, a classical/exact reference comparison, and an honest discussion of limitations.
- A short technical pitch explaining the approach **3-min demo** for evaluation (Google Drive or link).

Judging Rubric — 100 Points

CRITERION	WHAT TO ASSESS	PTS
A. Problem validity	Problem relevance within an eligible domain (5); clear optimization formulation (5); appropriate scope and feasibility (5); measurable objectives and constraints (5).	**20**
B. Meaningful quantum utilization	Correct mapping to a quantum representation (10); suitability and justification of algorithm (10); quantum computation materially used in the solution and awareness of quantum assumptions and limitations (5).	**25**
C. Implementation & validation	Working Qiskit implementation (10); interpretable outputs (5); valid classical/exact/analytical baseline (5); reproducibility of method and results (5).	**25**
D. Innovation & originality	Originality of the chosen problem within the domain (10); thoughtful adaptation, modeling, or technical extension (5).	**15**
E. Presentation & defense	Clarity of explanation (10); demonstration quality and answers to technical questions (5).	**15**
TOTAL	**Maximum available points**	**100**